ROYAL CRICKET STORE - TEST PLAN

White Box Testing (WBT) and Black Box Testing (BBT)

1.Introduction

Royal Cricket Store is an Ecommerce application Designed to help users to Shop all Cricket products like Bat , Ball, Cricket Pads, Helmet and all other Cricket Accessories At the Finger Tip of The User

Software Testing is performed to identify defects and verify that the application Behaves According to its requirements . This test Plan covers Both **White Box Testing** and **Black Box Testing** of The Royal Cricket Store Project

White Box Testing focuses on the internal program logic, statements, branches, conditions, loops, and execution paths. The WBT test cases for Royal Cricket Store cover seven major modules and contain 35 test cases.

Black Box Testing focuses on the externally visible behavior of the application. The tester provides inputs and verifies whether the obtained outputs match the expected results without considering the internal implementation.

2. Test Objectives

1.To verify that the **Royal Cricket Store** functions according to its specified requirements.
2.To identify and detect **errors, defects, and failures** in the software.
3.To verify the functionality of all major modules of the system.
4.To test the system using **White Box Testing (WBT)** and **Black Box Testing (BBT)** techniques.
5.To verify that the system provides **correct results for valid inputs**.
6.To ensure that the system **handles invalid inputs appropriately**.
7.To verify the interaction between different modules of the system.
8.To ensure that the software is **reliable, user-friendly, and efficient**.
9.To reduce the possibility of failures after deployment.
10.To ensure that the final system meets the **functional and technical requirements**

3.Test Scope

3.1 In-Scope Modules

The following Royal Cricket Store Modules are included in testing process

Module Id	Module Name
M01	Login
M02	Dashboard
M03	Products
M04	Cart
M05	Registration

3.2 Functionalities Covered

The following Functionalities will be tested :

- User Registration
- User login
- Password Validation
- JWT Authentication
- Product display
- Product Search
- Product details
- Shopping Cart
- User Logout
- Users Registration After Login
- Database error Handling

Out Of Scope

Testing of third-party services, external APIs, and payment gateway internals is not covered. Hardware, network infrastructure, and extreme-load performance testing are also outside the scope of this project.

4. Test Strategy

Two major testing approaches are used

4.1 White box testing strategy

White Box Testing is performed by examining the internal logic and execution structure of the Royal Cricket Store application

The following techniques are used:

Technique	Purpose
Statement Coverage	Verify that important executable statements are executed
Branch Coverage	Verify different branches of conditional statements
Condition Coverage	Verify individual logical conditions
Loop Testing	Verify execution of loops with different inputs
Path Coverage	Verify important execution paths
Basis Path Testing	Verify independent execution paths through the program

The Uploaded Royal Cricket Test Cases uses this techniques across authentication , dashboard, products, cart , payment, chatbot

4.2 Black Box Testing Strategy

Black Box Testing verifies the behavior of Royal Cricket Store from the user's perspective. The following approaches are used:

- Functional testing
- Input validation
- Valid input testing
- Invalid input testing
- Boundary-value-oriented testing
- Error handling testing
- Data validation testing

The internal source code is not considered while performing BBT. Only the input and resulting output are evaluated.

5. Test Resources

5.1 Hardware Resources

- Laptop
- Android Smart Phone
- Sufficient Storage for application
- Internet connection

5.2 Software Resources

- Royal Cricket Store Application
- Development environment used for project
- Database used
- Web browser
- Testing tools

5.3 Human Resources

Role	Responsibility
Tester1	Authentication testing
Tester2	Dashboard testing
Tester3	Product Testing

Tester4	**Cart Testing**

6.Test Schedule

The following Schedule can be used for the practical testing activity:

Activity	Duration
Test Plan Preparation	Day 1
Test Data Preparation	Day 1
WBT Test Case Preparation	Day 2
WBT Test Execution	Day 3–4
BBT Test Case Preparation	Day 5
BBT Test Execution	Day 6–7
Defect Recording	During testing
Defect Correction	Day 8
Retesting	Day 9
Regression Testing	Day 10
Final Test Report	Day 11

Tester No.	Tester Name
1	Avishkar Abasaheb Gorad
2	Shreyash Santosh Shejal
3	Tushar Kailas Wadmare

7. WBT TEST CASES

Module 1 - Login Module

Field	Description
Taste Case ID	TC_001
Title	Check if login works properly and redirects to dashboard
Description	Test if a user can log in with right username and password
Preconditions	The user must have registered and active account
Test Steps	1.Open The login page 2.Enter correct username 3.Enter correct password 4.Click Login Button
Test Data	Username: avishkargorad Password:password123
Expected Result	The user should redirected to the main dashboard
Actual Result	The user is redirected to the dashboard
Status	Pass
Postconditions	The user is logged in and sees the dashboard

Field	Description
Taste Case ID	TC_002
Title	Check login with empty username field
Description	Test if the system validates empty username input
Preconditions	Login page should be available
Test Steps	1. Open login page 2. Keep username field empty 3. Enter password 4. Click Login button
Test Data	Username: Empty Password: password123
Expected Result	System should show username required validation message
Actual Result	Username required message is displayed
Status	Pass
Postconditions	User remains on the login page

Module 2 - Dashboard

Field	Description
Taste Case ID	TC_DASHBOARD_003
Title	Check order summary display on dashboard
Description	Test if order-related information is displayed on the dashboard
Preconditions	User must have previous orders
Test Steps	1. Login into account 2. Open dashboard 3. Check order summary section
Test Data	Existing customer orders
Expected Result	Order summary should be displayed correctly
Actual Result	Order summary is displayed successfully
Status	Pass
Postconditions	User can view order information

Activate Windows

Module 3 - Products

Field	Description
Taste Case ID	TC_Product001
Title	Check if all products are displayed properly
Description	Test if the system displays all available cricket products from the database
Preconditions	Products must be available in the database
Test Steps	1. Open the Products page 2. Wait for products to load 3. Verify the displayed product list
Test Data	Product Database: Available products
Expected Result	All available products should be displayed on the products page
Actual Result	All products are displayed successfully
Status	Pass
Postconditions	User can view all available products

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Module 4 - Cart Module

Field	Description
Test Case ID	TC_CART_001
Title	Check if product can be added to cart
Description	Test if a user can successfully add a selected product to the shopping cart
Preconditions	User must be logged in and products should be available
Test Steps	1. Open Products page 2. Select a product 3. Click Add to Cart button
Test Data	Product Name: Cricket Bat
Expected Result	Selected product should be added to the cart
Actual Result	Product is added successfully to the cart
Status	Pass
Postconditions	Product appears in the user's cart

8.BBT TEST CASES

LOGIN MODULE:

Field	Description
Taste Case ID	TC_003
Title	Check login with invalid username and password
Description	Test if a user can log in with wrong username and password
Preconditions	The user must not have registered and should not have active account
Test Steps	1.Open The login page 2.Enter wrong username 3.Enter wrong password 4.Click Login Button
Test Data	Username: user_450 Password: 123456
Expected Result	The user should get a error message
Actual Result	The user has got the error message
Status	Pass
Postconditions	User remains on the login page

PRODUCT MODULE:

Field	Description
Taste Case ID	TC_product005
Title	Check product category filtering
Description	Test if products can be filtered according to category
Preconditions	Multiple product categories should be available
Test Steps	1. Open Products page 2. Select product category 3. Check displayed products
Test Data	Category: Cricket Equipment
Expected Result	Only selected category products should be displayed
Actual Result	Filtered products are displayed correctly
Status	Pass
Postconditions	User can browse products according to category

CART MODULE:

Field	Description
Test Case ID	TC_CART_004
Title	Check adding multiple products to cart
Description	Test if multiple products can be added to the cart
Preconditions	Multiple products should be available
Test Steps	1. Open Products page 2. Select multiple products 3. Click Add to Cart
Test Data	Products: Cricket Bat, Cricket Ball, Gloves
Expected Result	All selected products should be added to the cart
Actual Result	Multiple products are added successfully
Status	Pass
Postconditions	Cart contains all selected products

CONCLUSION

The Test Plan provides a systematic approach for verifying the functionality, reliability, and quality of the Royal Cricket Store. By applying appropriate White Box and Black Box Testing techniques, the system can be effectively evaluated, defects can be identified, and the software can be improved to meet its specified requirements.